Coding & Solution

Date	01 JULY 2026
Team ID	xxxxx
Project Name	Opticrop:Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This document presents the implementation details of the OptiCrop system. The solution demonstrates clean coding practices, modular implementation, and successful integration of Machine Learning with a Flask web application for crop recommendation.

Solution Summary

Field	***Details***
Repository Link / URL	Github link : https://github.com/hasinibadviti/OptiCrop-Smart-Agricultural-Production-Optimization-Engine/tree/main Live url : http://127.0.0.1:5000
Programming Language(s)	PYTHON,HTML,CSS,JAVA SCRIPT
Framework(s) Used	FLASK
Key Features Implemented	1.Crop Recommendation Prediction using Machine Learning. 2.Data preprocessing and feature scaling using Pandas and Scikit-learn. 3.Interactive Flask web interface for user input and prediction. 4.Real-time crop recommendation based on soil and environmental parameters. 5.Model training, evaluation, and prediction using trained ML model.

Pending / Incomplete Features	1. Weather API integration for live climate data. 2. User authentication and prediction history. 3. Mobile responsive UI and cloud deployment.
Setup / Run Instructions	1. Create a virtual environment. 2. Install dependencies: pip install -r requirements.txt 3. Run the Flask application: python app.py 4. Open:http://127.0.0.1:5000

Code Quality Checklist

S.No	*Criteria*	*Status (Yes / No)*
1	Code is modular and organized into functions / classes	YES
2	Meaningful variable and function names are used	YES
3	Code includes comments / documentation where necessary	YES
4	Error handling is implemented for critical operations	YES
5	The application runs without critical errors	YES
6	Code is committed to a version control repository	YES

Additional Notes / Comments:

Machine Learning pipeline includes preprocessing, model training, and prediction using Scikit-learn.

Input validation and exception handling improve application reliability.